

RHEL Lightspeed

Version 1.0.0, 2026-06-16

Home

Introduction

Course Title: RHEL Lightspeed

Description: Course description FIXME

Duration: FIXME hours

Topics covered in this course

- Introducing RHEL Lightspeed
- Using the Command-line Assistant
- Hands-on Lab

Prerequisites

This course assumes that you have the following prior experience:

- Course or experience...
- Course or experience...
- Course or experience...

Lab Environment

FIXME: Instructions for accessing the lab environment for this hands-on based training.

Introducing RHEL Lightspeed

RHEL Lightspeed is an intelligent assistant designed to simplify system administration by enabling users to interact with RHEL using plain, natural language.

Key functionalities and benefits include:

- **Accessible Management:** Streamlines RHEL operations for administrators of all experience levels by reducing reliance on complex command syntax.
- **Intelligent Interaction:** Allows users to query the system, troubleshoot issues, and decipher log entries through intuitive, conversational prompts.
- **Enhanced Productivity:** Bridges the gap between technical requirements and execution, making RHEL environments easier to navigate and manage.

Introducing RHEL Lightspeed

RHEL Lightspeed is an intelligent, AI-powered tool integrated into the Red Hat Enterprise Linux (RHEL) ecosystem. It is designed to act as a force multiplier for system administrators, bridging the gap between complex manual operations and natural language interaction.

Intelligent Functionalities

At its core, RHEL Lightspeed leverages deep integration with Red Hat's vast repository of technical documentation, including Knowledge Centered Service (KCS) articles and official product documentation. Its primary intelligent functionalities include:

- **Natural Language Processing (NLP):** RHEL Lightspeed translates plain-language inquiries into actionable technical tasks or explanations, removing the requirement to memorize complex command-line syntax for every operation.
- **Context-Aware Troubleshooting:** The assistant can analyze system outputs, error messages, and log entries to provide root-cause analysis and remediation suggestions, effectively reducing the time spent navigating traditional troubleshooting workflows.
- **Knowledge Synthesis:** By pulling from verified Red Hat resources, the assistant provides guidance that aligns with industry best practices, ensuring that recommendations are both accurate and secure.
- **Interactive Workflow Support:** Beyond answering static questions, RHEL Lightspeed supports iterative problem-solving. You can engage in a multi-turn conversation to diagnose issues, implement new features, or explore RHEL configuration options.

Benefits for System Administrators

RHEL Lightspeed is engineered to be inclusive, offering specific value regardless of your tenure as an administrator:

User Profile	Benefit
Less Experienced Users	Lowers the barrier to entry by providing guided, step-by-step instructions for tasks that might otherwise require deep knowledge of manual man pages or complex command flags.
Experienced Users	Increases operational velocity by providing instant access to specific KCS information, troubleshooting logic, and complex syntax, allowing for faster resolution of routine and edge-case issues.

Accessibility and Management

The integration of RHEL Lightspeed into the command-line interface makes advanced intelligence accessible directly within the administrator's workspace. This architectural decision ensures that:

- **Reduced Context Switching:** Administrators can remain within their terminal environments while accessing expert-level guidance, rather than leaving the system to search external web documentation.
- **Standardized Management:** Because the assistant draws from consistent, authoritative Red Hat data, it promotes standardization across diverse RHEL environments, helping teams maintain compliance and best-practice configurations even when dealing with unfamiliar RHEL features.

By transforming the RHEL command-line interface from a static environment into an interactive, intelligent partner, RHEL Lightspeed empowers administrators to manage, maintain, and scale their infrastructure with greater confidence and efficiency.

Benefits for system administrators of all experience levels

RHEL Lightspeed is designed to bridge the gap between technical complexity and operational efficiency. By integrating generative AI into the Red Hat Enterprise Linux (RHEL) ecosystem, it provides a supportive layer that adapts to the user's proficiency, ensuring that both novices and seasoned experts can manage their environments more effectively.

Empowering Less Experienced Administrators

For administrators newer to the RHEL ecosystem, the command-line interface can often feel daunting due to the vast array of flags, syntax requirements, and subsystem interdependencies. RHEL Lightspeed lowers this barrier to entry by:

- **Translating Natural Language to Commands:** Users can describe the desired outcome—such as "create a new user with sudo privileges and a custom shell"—without needing to memorize complex `useradd` syntax.
- **Contextual Learning:** By converting plain-language requests into executable commands, the assistant acts as a "teach-by-doing" tool, helping administrators build technical intuition over time.

- **Reduced Cognitive Load:** Instead of spending time searching through extensive documentation (man pages or Knowledgebase articles), users receive direct, accurate answers relevant to their specific system state.

Enhancing Productivity for Experienced Professionals

For seasoned system administrators, RHEL Lightspeed serves as a force multiplier, shifting the focus from manual syntax construction to high-level architecture and troubleshooting. Key benefits include:

- **Acceleration of Routine Tasks:** Experienced users can bypass the "lookup phase" for complex or infrequently used commands, allowing for rapid configuration and deployment.
- **Advanced Troubleshooting:** When facing cryptic log files or complex SELinux denials, RHEL Lightspeed can ingest, parse, and provide an analysis of the output. This significantly reduces the time required for root-cause analysis.
- **Consistency and Best Practices:** The assistant helps ensure that tasks are performed according to RHEL-recommended practices, reducing the risk of "configuration drift" or manual errors that can occur during repetitive tasks.

Shared Advantages

Regardless of experience level, RHEL Lightspeed provides universal benefits that streamline daily operations:

- **Operational Efficiency:** By minimizing the need to switch context between the terminal and external documentation, administrators stay in the "flow state" of their work.
- **Accessibility:** Complex system management tasks become more accessible, as the barrier of memorizing specific command-line switches is removed.
- **Unified Support:** Whether debugging a failing `sshd` service or optimizing system performance, RHEL Lightspeed provides a consistent interface for interaction, making it a reliable partner in maintaining the health and security of the RHEL environment.



RHEL Lightspeed is intended to assist and augment the administrator's expertise. While it provides powerful, actionable guidance, it is important to review the suggested commands before execution, especially in production environments.

Accessibility and Ease of Management in RHEL Environments

Managing complex enterprise infrastructure often involves navigating vast documentation, complex syntax, and intricate troubleshooting steps. RHEL Lightspeed transforms this experience by lowering the barrier to entry, ensuring that RHEL environments remain accessible and manageable for administrators across all skill levels.

Bridging the Experience Gap

The primary challenge in managing RHEL at scale is the learning curve associated with its deep, robust feature set. RHEL Lightspeed serves as an intelligent bridge, offering several key advantages for system administrators:

- **Natural Language Interaction:** Users are no longer restricted to memorizing complex flags, arguments, or obscure syntax. By using natural language, administrators can express the **intent** of an action—such as "How do I configure a static IP on this interface?"—rather than hunting for the exact command-line parameters.
- **Contextual Assistance:** By integrating Red Hat’s official product documentation and the Red Hat Knowledgebase directly into the command-line interface, RHEL Lightspeed provides expert-level guidance that is context-aware. This ensures that the advice provided is accurate to the RHEL version being managed.
- **Reduced Cognitive Load:** Experienced administrators often encounter "syntax fatigue" when working across multiple disparate systems. RHEL Lightspeed allows for a shift in focus from "how to write the command" to "what needs to be achieved," streamlining daily operations.

Enhancing Operational Efficiency

Accessibility, in the context of RHEL, means more than just ease of use; it implies a more reliable and efficient operational workflow. RHEL Lightspeed enhances management through:

Feature	Benefit to Management
Simplified Troubleshooting	Translates complex log outputs and system errors into plain language explanations, allowing for faster root-cause analysis.
Task Acceleration	Automates the drafting of configuration commands, reducing the time spent referencing manuals during routine maintenance.
Continuous Learning	Provides immediate explanations for executed commands, turning routine tasks into educational opportunities for junior staff.

By integrating these intelligent functionalities, RHEL Lightspeed ensures that the power of Red Hat Enterprise Linux is accessible to everyone, from those performing their first installation to seasoned architects managing global data centers. This democratization of expertise ensures that system configuration and maintenance remain consistent, accurate, and highly efficient.

Using the Command-line Assistant

Using the Command-line Assistant

The RHEL Lightspeed command-line assistant allows you to interact with your system using plain language, simplifying complex management and troubleshooting tasks. By converting natural language queries into RHEL commands, it helps users of all experience levels perform administrative duties more efficiently.

Key features include:

- **Natural Language Interaction:** Ask questions about RHEL configurations or system tasks without needing to recall complex command syntax.
- **Troubleshooting and Analysis:** Use the assistant to diagnose issues, such as service failures, or to decipher intricate log entries and command outputs.
- **Workflow Optimization:** Transition from manual, multi-step procedures to streamlined queries for common operations, such as file searches or service status checks.



Do not input personal, confidential, or business-sensitive data into the assistant. All queries are sent for AI inference, and you are responsible for ensuring that no sensitive system or personal information is included in your prompts.

Interacting with the command-line assistant using plain language

The RHEL Lightspeed command-line assistant is designed to bridge the gap between complex administrative tasks and natural language interaction. By leveraging Large Language Models (LLMs), it allows you to describe your intent in plain English rather than memorizing complex command-line syntax and flags.

The Interaction Workflow

When you interact with the command-line assistant, your input undergoes a structured process to ensure the output is contextually relevant and technically accurate.

1. **Input Submission:** You provide a prompt using natural language.
2. **Backend Processing:** The assistant captures your query and searches internal, verified Red Hat knowledge bases to retrieve relevant documentation and configuration guidance.
3. **AI Inference:** Your query, combined with the retrieved knowledge and system instructions, is sent to the configured LLM provider to generate an accurate technical response.
4. **Output Delivery:** The assistant provides the solution, which may include commands, explanations, or troubleshooting steps directly in your terminal.

Best Practices for Prompting

To get the most out of the assistant, structure your requests to be clear and specific. Instead of providing broad, vague statements, frame your queries as direct operational tasks.

- **Be specific:** Include the component or service you are working with (e.g., `sshd`, `firewalld`, `dnf`).
- **Define the goal:** Clearly state the outcome you are looking for, such as troubleshooting a service failure or searching for specific file patterns.
- **Use action-oriented language:** Start with verbs like "troubleshoot," "list," "find," or "configure."

Examples of Natural Language Queries

Instead of searching for complex manual pages or navigating documentation portals, you can use the `c` command to ask questions directly:

- **Troubleshooting:** `$ c "how to troubleshoot sshd failing to start"`
- **File Operations:** `$ c "how do I find all the files in the /etc that have been modified in the last hour"`

Data Privacy and Security Warning



Security Notice: The command-line assistant is not intended to process sensitive data. Do not include any information in your prompts that you would not want shared with an external LLM provider.

Avoid entering the following types of data: * **Personal Information:** Usernames, email addresses, or personal contact details. * **Business Sensitive Information:** Proprietary application code, internal project names, or roadmap details. * **Confidential Information:** Passwords, private keys, or API tokens. * **System Data Information:** Specific IP addresses, hostnames, or unique infrastructure metadata that could identify your specific environment.

By using the assistant, you acknowledge that Red Hat may use the message exchanges to improve the functionality of the service. Always prioritize system security by sanitizing your queries before submission.

Answering RHEL-related questions

The RHEL Lightspeed command-line assistant acts as an intelligent bridge between you and the vast ecosystem of Red Hat documentation. By leveraging generative AI trained on official RHEL product documentation and the Red Hat Knowledgebase, the assistant allows you to retrieve technical information, best practices, and configuration guidance without leaving your terminal.

Understanding the Interaction Model

The command-line assistant is designed to interpret plain language queries. Instead of searching through manual pages or navigating web-based portals, you can pose direct questions to the

system. This is particularly useful for:

- **Concept Clarification:** Understanding high-level RHEL features or subsystems.
- **Best Practice Guidance:** Determining the recommended way to perform common administrative tasks.
- **Documentation Retrieval:** Obtaining quick summaries of commands or configuration file parameters.

Syntax and Usage

To interact with the assistant, use the `c` command followed by your query enclosed in quotation marks. Using quotation marks ensures that your full query is passed correctly to the AI model, preventing shell interpretation of special characters or spaces.

```
$ c "your question here"
```

Examples of Effective Queries

To get the most accurate information, structure your questions clearly. Here are a few examples of how to query the assistant:

- **Conceptual:** `c "What is the purpose of the tuned daemon in RHEL?"`
- **Configuration:** `c "How do I add a persistent static route in RHEL 10?"`
- **Tooling:** `c "What is the difference between dnf and yum?"`

Best Practices for Querying

While the assistant is highly capable, the quality of the answer often depends on the specificity of the question. Follow these guidelines for better results:

- **Be Specific:** Instead of "How do I manage services?", try "How do I enable and start the httpd service in RHEL?"
- **Include Context:** If you are asking about a specific version, include it in your query (e.g., "In RHEL 10...").
- **Iterate:** If the initial response is too broad, follow up with a more focused question to narrow down the technical details.



The RHEL Lightspeed command-line assistant provides information based on verified Red Hat sources. Always review the output provided, especially when executing commands that modify system configurations, to ensure they align with your specific environment's requirements.

Troubleshooting system issues

Troubleshooting system issues with RHEL Lightspeed

RHEL Lightspeed transforms the traditional troubleshooting workflow by acting as an intelligent co-pilot. Instead of manually parsing man pages or searching through external documentation, you can use the command-line assistant to diagnose system behaviors, analyze logs, and identify remediation steps using natural language.

Using the Command-line Assistant for Diagnostics

The command-line assistant is designed to understand context-aware queries. By using the **c** command prefix, you can initiate a conversation with the AI to troubleshoot services, network configurations, and security policies.

Best Practices for Queries

- **Be Specific:** Provide the service name or the specific error code you are encountering.
- **State the Symptom:** Clearly describe what is failing (e.g., "I cannot access my server with SSH").
- **Request Remediation:** Ask for a list of troubleshooting steps to receive a structured plan rather than a single command.

Common Troubleshooting Scenarios

The assistant is capable of assisting with a wide variety of RHEL-specific issues, such as:

- **Service Failures:** Querying why a daemon, like **sssd** or **sshd**, fails to start.
- **Network Connectivity:** Identifying misconfigurations in firewall settings or network interfaces.
- **Boot Management:** Determining how to boot into a specific kernel version during maintenance.
- **File Auditing:** Searching for system changes, such as identifying files in **/etc** modified within a specific timeframe.



When troubleshooting, always verify the suggestions provided by the assistant against your organization's security policies before executing them on production systems.

Hands-on Lab: Analyzing Logs and Troubleshooting

In this lab, you will use the RHEL Lightspeed assistant to resolve a simulated service issue.

Prerequisites

- Ensure that the RHEL Lightspeed command-line assistant is installed and configured on your host.
- Verify that you have authenticated with the required Red Hat credentials.

Step 1: Querying a Service Failure

If the **sssd** service fails to start, you can request an immediate diagnostic path.

1. Open your terminal.
2. Enter the following command to ask the assistant for troubleshooting steps:

```
$ c "I am failing to start sssd process, what should I check first?"
```

3. Review the assistant's output, which will typically include checking status commands (`systemctl status sssd`) and log file locations (`/var/log/sss/`).

Step 2: Troubleshooting SELinux

If you encounter "Permission Denied" errors, the assistant can help identify if SELinux is the root cause.

1. Use the assistant to query common SELinux troubleshooting steps:

```
$ c "I am getting permission denied errors, how do I check if SELinux is blocking my application?"
```

2. Follow the suggested steps provided by the assistant, which may involve checking `ausearch` or `sealert` outputs.

Step 3: Finding Recently Modified Files

Use the assistant to help construct complex commands for administrative tasks, such as tracking changes.

1. Query the assistant for a command to audit configuration changes:

```
$ c "how do I find all the files in the /etc that have been modified in the last hour"
```

2. Execute the suggested `find` command provided by the assistant and verify the results against your `/etc` directory.

Deciphering log entries and system outputs

System administrators frequently encounter complex log files and cryptic command outputs that can be difficult to interpret, especially during high-pressure troubleshooting scenarios. RHEL Lightspeed streamlines this process by acting as an intelligent interface that translates raw, machine-readable data into actionable human-readable insights.

Understanding Logs with RHEL Lightspeed

Rather than manually parsing through hundreds of lines of text or scouring documentation to understand error codes, you can leverage the `c` command-line assistant to interpret system outputs directly in your terminal. This capability significantly reduces the time spent on root cause analysis.

Key Advantages

- **Contextual Interpretation:** The assistant analyzes the specific output provided, identifying patterns and common failure signatures.
- **Plain Language Explanations:** It converts technical jargon into clear summaries, explaining what a specific error means for the service or system.
- **Guided Remediation:** Beyond explanation, the assistant often suggests specific commands or configuration changes to resolve the identified issue.

Hands-on Lab: Analyzing System Logs

In this lab, you will learn how to feed system logs directly into the RHEL Lightspeed assistant to identify and resolve service errors.

Prerequisites

- Ensure the RHEL Lightspeed command-line assistant is configured and authenticated in your environment.

Activity 1: Interpreting Service Failures

In this exercise, you will investigate a failing `sshd.service` by piping its output to the assistant for diagnosis.

1. **Generate and Pipe Log Data:** Use `journalctl` to pull the recent logs for the SSH service, filter for the last 30 lines, and pipe the output to the assistant with a specific request for interpretation.

```
$ journalctl -xau sshd.service | tail -n 30 | c 'here are the logs, please help me understand this'
```

2. **Review the Output:** The assistant will process the piped log data. Observe how it identifies potential misconfigurations or syntax errors within the configuration files based on the log trail.
3. **Request Remediation:** Once the assistant identifies the error (e.g., a typing error in the configuration file), ask it to generate the necessary fix.

```
$ c 'how do I fix the error identified in these logs?'
```

Activity 2: Investigating Arbitrary Log Files

Sometimes you may need to analyze logs that are stored in flat files rather than the system journal.

1. **Redirecting File Content:** If you have a log file that is reporting errors, use the `cat` command to redirect its content to the assistant.

```
$ cat /var/log/my_application.log | c
```

2. **Refining Your Query:** If the initial output is too broad, provide a specific query alongside the redirected output to narrow the assistant's focus.

```
$ cat /var/log/my_application.log | c 'Identify the specific error causing the application crash and suggest a troubleshooting step.'
```

Troubleshooting Tips

- **Be Specific:** When piping logs, always include a brief prompt (e.g., "explain this error" or "what is the cause of this failure?") to ensure the assistant focuses on the relevant parts of the log.
- **Contextual History:** Remember that if you need to reference the output of a command you just ran, you can use the `-w` flag. For example, `c -w 1 "explain this"` references the output of the single immediately preceding command.
- **Quality of Data:** Ensure the piped data is relevant to the issue. Large log dumps may contain noise; use filters like `grep` or `tail` before piping to the assistant to improve the accuracy of the generated response.

Transitioning from complex commands to natural language queries

The RHEL Lightspeed command-line assistant serves as a bridge between the traditional, high-precision world of Linux command-line syntax and the intuitive, flexible nature of human language. By leveraging generative AI, you can shift your focus from memorizing complex flag combinations and syntax structures to defining the **intent** of your system management tasks.

The Shift: From Syntax to Intent

Traditional RHEL administration requires significant cognitive load to construct commands with the correct arguments, flags, and pipes. Transitioning to natural language queries allows you to interact with the system in a more conversational manner.

Instead of needing to recall the exact syntax for a multi-stage command, you describe the desired state or the problem you are facing. The RHEL Lightspeed assistant processes this request by:

1. **Parsing Intent:** Identifying the core task (e.g., checking service status, filtering logs, or adjusting permissions).
2. **Contextual Mapping:** Utilizing Red Hat product documentation and the Red Hat Knowledgebase to translate your intent into actionable commands.
3. **Syntactic Precision:** Generating the exact, valid command structure required for the current version of RHEL.

How It Works: The Input/Output Flow

When you transition from typing complex commands to using the assistant, you are interacting with a backend pipeline designed to ensure accuracy:

1. **Input:** You enter your question or task in natural language.
2. **Processing:** The command-line assistant searches for relevant technical knowledge specific to your query.
3. **Inference:** Your query, combined with the retrieved knowledge and system instructions, is sent to the Large Language Model (LLM) for inference.
4. **Output:** The assistant provides a clear, actionable response, which may include the precise command you need to run.



Data Privacy Notice: When transitioning to natural language queries, ensure that you do not include personal information or sensitive company data in your input. The command-line assistant is not intended to process personal information.

Practical Example: Simplifying Syntax

Consider a scenario where you need to analyze a system log to find errors related to a specific service.

- **Traditional Method:** You would need to remember the correct syntax for `journalctl`, including appropriate flags for filtering by priority and unit, such as: `journalctl -u httpd.service --since "1 hour ago" --priority=err`
- **Natural Language Query:** You can simply ask the assistant: **"Show me the error logs for the Apache web server from the last hour."**

The assistant translates this intent, references the relevant RHEL documentation, and provides the correct command along with an explanation of what the output means. This transition not only increases efficiency but also reduces the likelihood of syntax errors that occur when manually constructing complex command strings.

Troubleshooting Through Natural Language

Beyond executing tasks, you can use natural language to troubleshoot by redirecting log file outputs. If you have a log file that contains a cryptic error, you do not need to manually parse it or search for individual patterns. You can redirect the log content directly to the command-line assistant, enabling you to ask: **"Explain what this error means and how I should resolve it."**

This approach transforms the troubleshooting process from a manual, time-consuming research task into an interactive, expert-guided conversation.

hands-on-lab.adoc

Hands-on Lab

The hands-on lab provides a practical environment to master the RHEL Lightspeed command-line assistant. Participants will engage in the following exercises:

- **Configuration:** Setting up the command-line assistant for immediate use within the RHEL environment.
- **Natural Language Execution:** Practicing the transition from complex command syntax to intuitive natural language prompts for system administration tasks.
- **Troubleshooting Scenarios:** Utilizing AI-driven analysis to interpret system logs and resolve common issues effectively.

Configuring the RHEL Lightspeed command-line assistant

The RHEL Lightspeed command-line assistant transforms how you interact with your RHEL environment by bridging the gap between natural language queries and complex system administration tasks. Proper configuration is the foundation for leveraging these intelligent features effectively.

Prerequisites

- A supported RHEL architecture (x86_64, aarch64, s390x, or ppc64).
- Access to official Red Hat RHEL repositories.
- Administrative privileges (**root** or **sudo** access) on the target RHEL system.

Installation Guidelines

To ensure system stability and supportability, you must install the assistant exclusively through the official RHEL repositories.



Do not use **pip install** to install the command-line assistant. Using third-party package managers or **pip** is not supported by Red Hat and may result in dependency conflicts or security vulnerabilities.

Installation Procedure

1. **Update your repository metadata** to ensure you are fetching the latest available version:

```
sudo dnf update
```

2. **Install the package** using the system package manager:


```
sudo dnf install rhel-lightspeed-assistant
```

3. **Verify the installation** by checking the version and confirming the command is available in your `$PATH`:

```
rhel-lightspeed --version
```

Hands-on Lab: Initializing the Assistant

In this lab, you will prepare the RHEL Lightspeed assistant for first-time use.

Step 1: Authentication and Setup

Depending on your organizational setup, the assistant requires authentication to connect to the RHEL Lightspeed service.

1. Run the initialization command to trigger the setup wizard:

```
rhel-lightspeed configure
```

2. Follow the on-screen prompts. You may be asked to authenticate via a browser-based flow or by providing a service token provided by your Red Hat organization.

Step 2: Validating Connectivity

Once configured, test the connection to ensure the assistant can communicate with the backend intelligence service.

1. Execute a simple verification query:

```
rhel-lightspeed "How do I check my system uptime?"
```

2. If configured correctly, the assistant will return a plain-language explanation followed by the relevant command (e.g., `uptime`).

Step 3: Troubleshooting Configuration Issues

If the assistant fails to respond or reports a connection error:

- **Check Network Proxy:** If your environment uses a proxy, ensure the `http_proxy` and `https_proxy` environment variables are correctly set in your shell session.
- **Verify Repository Access:** Ensure your system is correctly registered to the Red Hat Subscription Manager using `subscription-manager status`.
- **Check Logs:** If the initialization fails, review the logs located at `/var/log/rhel-lightspeed/` for specific error codes regarding API authentication or network timeouts.

Executing system tasks using natural language prompts

The RHEL Lightspeed command-line assistant bridges the gap between complex system administration tasks and human-readable intent. By leveraging natural language processing, you can offload the burden of memorizing intricate syntax, man pages, and obscure flag combinations, allowing you to focus on the operational outcome rather than the command structure.

Understanding the Interaction Model

When you submit a prompt to the assistant, you are not simply running a script; you are initiating a multi-stage reasoning process. The backend infrastructure:

1. Receives your natural language request.
2. Identifies relevant technical documentation and RHEL-specific knowledge bases.
3. Packages your query alongside context for the configured Large Language Model (LLM) provider.
4. Returns a response that bridges the gap between your intent and the required RHEL command.

Best Practices for Prompting

To achieve the best results when executing tasks, follow these guidelines:

- **Be Specific:** Instead of asking "how do I fix the network," provide context: "How do I restart the NetworkManager service and check its status?"
- **Focus on Intent:** State what you want the system to achieve. For example, "Show me all files modified in `/var/log` within the last hour."
- **Safety First:** Remember that the tool is a guide. Always review the suggested commands before executing them on production systems.



Do not include sensitive data in your prompts. Avoid entering:

- * Personal or business-sensitive information.
- * Confidential system configuration details or passwords.
- * Data that should not be shared with external LLM providers.

Hands-on Activity: Executing System Tasks

In this lab, you will use the command-line assistant to perform routine administrative tasks using natural language.

Prerequisites

- Ensure the RHEL Lightspeed command-line assistant is configured in your environment.
- Authenticate as a user with sufficient permissions to query system information.

Step 1: Querying for System Tasks

Use the command-line assistant to generate commands for common maintenance tasks.

1. Open your terminal.
2. Input a request to find specific configuration files:

```
$ c "how do I find all the files in the /etc directory that have been modified in the last 60 minutes?"
```

3. Review the output provided by the assistant. It should return a command structure similar to `find /etc -mmin -60`.

Step 2: Troubleshooting with Natural Language

Use the assistant to diagnose a simulated service issue.

1. Input a request to troubleshoot a common service failure:

```
$ c "how to troubleshoot sshd failing to start?"
```

2. Analyze the assistant's response. It will likely suggest commands such as `systemctl status sshd` or `journalctl -u sshd`.
3. Execute the suggested troubleshooting commands to gather information about the service state.

Step 3: Transitioning to Execution

Once you are comfortable with the output:

1. Verify the suggested command is safe for your current environment.
2. Copy or execute the command provided by the assistant to resolve the task.
3. If the assistant suggests a multi-step process, execute each step sequentially, verifying the output of each command before moving to the next.



If the assistant's suggestion is not precise, refine your prompt. For example, if you require the output in a specific format, append "output the result as a one-line command" to your prompt.

Analyzing Logs and Troubleshooting Scenarios with RHEL Lightspeed

One of the most powerful features of RHEL Lightspeed is its ability to interpret complex system outputs and logs. By offloading the initial analysis of error messages and configuration files to the AI-driven command-line assistant, you can significantly reduce the time spent identifying the root cause of system issues.

Analyzing System Logs

System logs often contain dense, cryptic information. RHEL Lightspeed simplifies this by allowing you to redirect log outputs directly into the assistant for an immediate, plain-language explanation.

To analyze a log file, you can pipe the contents of the file into the `c` command. This enables the

assistant to process the raw text and provide context, potential causes, and suggested remediation steps.



Ensure you have appropriate permissions to read the log files before piping them to the command-line assistant.

Troubleshooting Scenarios

Beyond interpreting logs, you can use the assistant to walk through interactive troubleshooting sessions. Whether you are dealing with service failures, network connectivity issues, or kernel configuration, you can engage with the assistant using natural language.

Effective Querying Patterns

To get the most relevant guidance from the assistant, frame your queries by providing: * **The symptom:** State what is failing (e.g., "SSSD fails to start"). * **The context:** Include relevant environment details or recent changes. * **The goal:** Specify what you are trying to achieve (e.g., "I need to find the configuration error").

For example, instead of searching through documentation, you can ask: `$ c "I am failing to start the sssd process, analyze the last 50 lines of /var/log/sss/sssd.log and suggest a fix."`

Hands-on Lab: Troubleshooting with RHEL Lightspeed

In this lab, you will simulate a common troubleshooting scenario by feeding log data into the RHEL Lightspeed assistant.

Prerequisites

- A system running RHEL with the RHEL Lightspeed command-line assistant configured.

Step 1: Redirecting Log Output

In this step, we will provide the assistant with log data to identify an error.

1. Open your terminal.
2. Suppose you suspect an issue in your `sssd` service. Redirect the log output to the assistant:

```
$ cat /var/log/sss/sssd.log | c "Analyze this log for errors and explain why the service failed to start"
```

3. Review the assistant's output. It will identify the specific error code or configuration syntax issue within the provided log snippet.

Step 2: Interactive Troubleshooting

Once the assistant identifies the issue, use it to refine your troubleshooting steps.

1. Based on the output from Step 1, ask the assistant for a correction:

```
$ c "The log indicates a syntax error in /etc/sss/sss.conf. How do I validate this file for errors?"
```

2. Execute the suggested command provided by the assistant to verify the configuration.
3. If the command returns another error, pipe that specific error back into the assistant:

```
$ <command_that_failed> 2>&1 | c "Explain this error and provide the correct syntax"
```

Step 3: Cleanup and Verification

1. Once the configuration is fixed, restart the service as suggested by the assistant.
2. Confirm the service is running:

```
$ systemctl status sssd
```

By integrating the assistant into your workflow, you move from manually parsing files to guided, intelligent resolution.

Appendix

FIXME: Content for Appendix...

Resources

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